

Lab - 4 : Constrained growth beyond the logistic model

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In this lab, we analyzed the Gompertz equation for the modelling of tumour growth. We also analyzed the Allee effect which models high growth rate of population when the initial size of the population has an intermediate value.

A. Model

The Gompertz equation models tumour growth as

$$\dot{x} = -ax \ln(bx) \quad (1)$$

Where $a, b > 0$.

Rescale $X = x/b^{-1}$ and $T = at$,

$$\dot{X} = -X \ln X \quad (2)$$

The integral solution of above equation is,

$$X = e^{[\ln(X_{in})e^{-T}]} \quad (3)$$

B. Results

Fig. 1 Plot of $\dot{x}$ vs x .

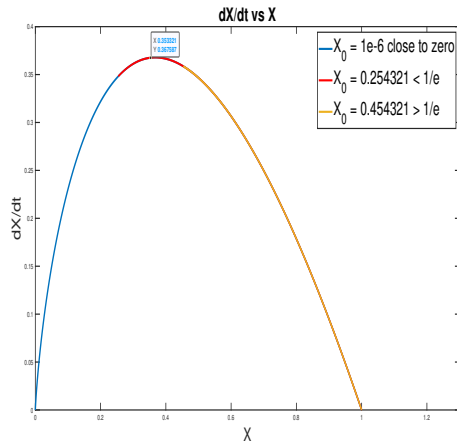


FIG. 1: Here the initial values of x are 10^{-6} , 0.254321 and 0.454321 $\Delta x = 0.01unit$

Fig. 2 Integral solution of $X(t)$ and shows comparison between the analytical solution and the numerical solution.

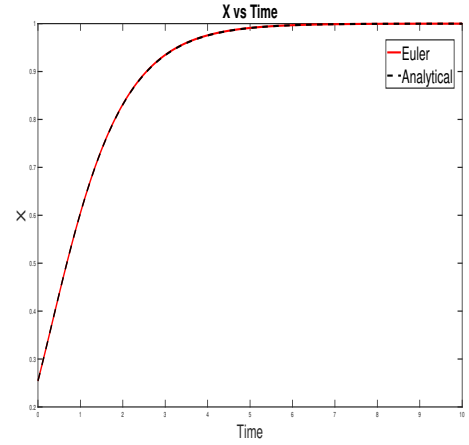


FIG. 2: Here the initial value of $x(0) = 0.1838$. $\Delta t = 0.01unit$

Fig. 3 shows relative error between the analytical solution and the numerical solution .

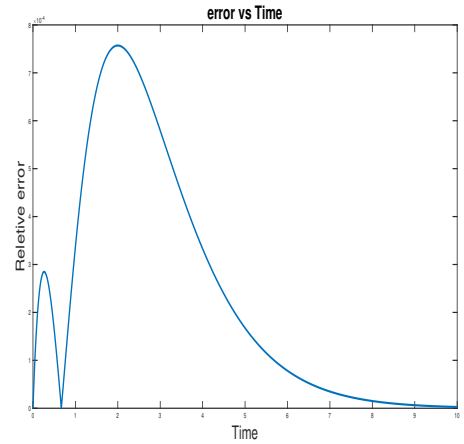


FIG. 3: The graph has zero value at $t=0$ and 1.175. $\Delta t = 0.01unit$

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C. Model

The Allee effect models high growth rate of a population when the initial population size has an intermediate value. The model equation is,

$$\dot{x} = x[r - a(x - b)^2] \quad (4)$$

Where $a, b, r > 0$. The model is effective only when $r < ab^2$.

D. Results

Fig. 4 Plot of $\dot{x}$ vs x .

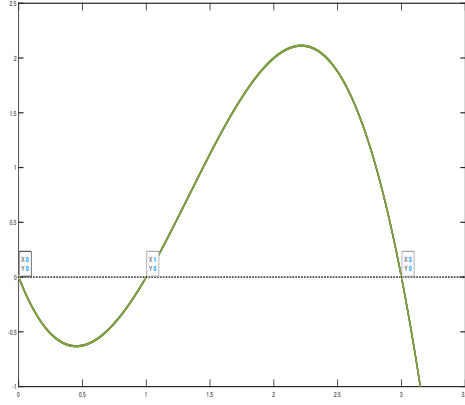


FIG. 4: Here $a=1, b=2, r=1$. Graph value becomes zero at $t=0, 1$ and 3 units. $\Delta t = 0.01 \text{ unit}$

Fig. 5 Integral solution of $x(t)$.

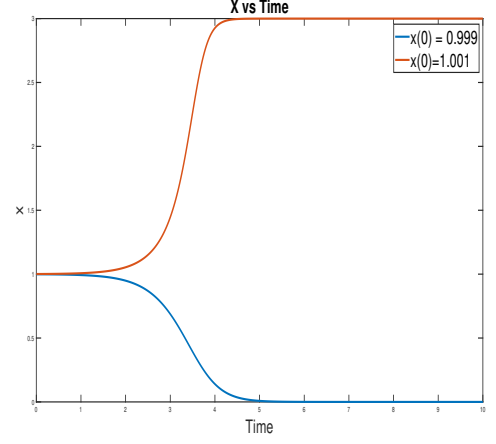


FIG. 5: Thw two initial values taken are 0.999 and 1.001. $\Delta x = 0.01 \text{ unit}$

I. CONCLUSION

- The rescaled Gompertz equation simplifies the tumor growth model to a parameter-free differential equation, making it easier to analyze and understand.
- In the Gompertz model, the plot of $\dot{X}$ VS X reveals the behavior of the tumor growth rate for different initial conditions, providing insights into the system's dynamics.
- The Allee effect is observed in the population growth model, where an intermediate initial population size leads to higher growth rates. The limiting values of x as $t \rightarrow \infty$ provide insights into the long-term behavior of the population under Allee effect conditions.